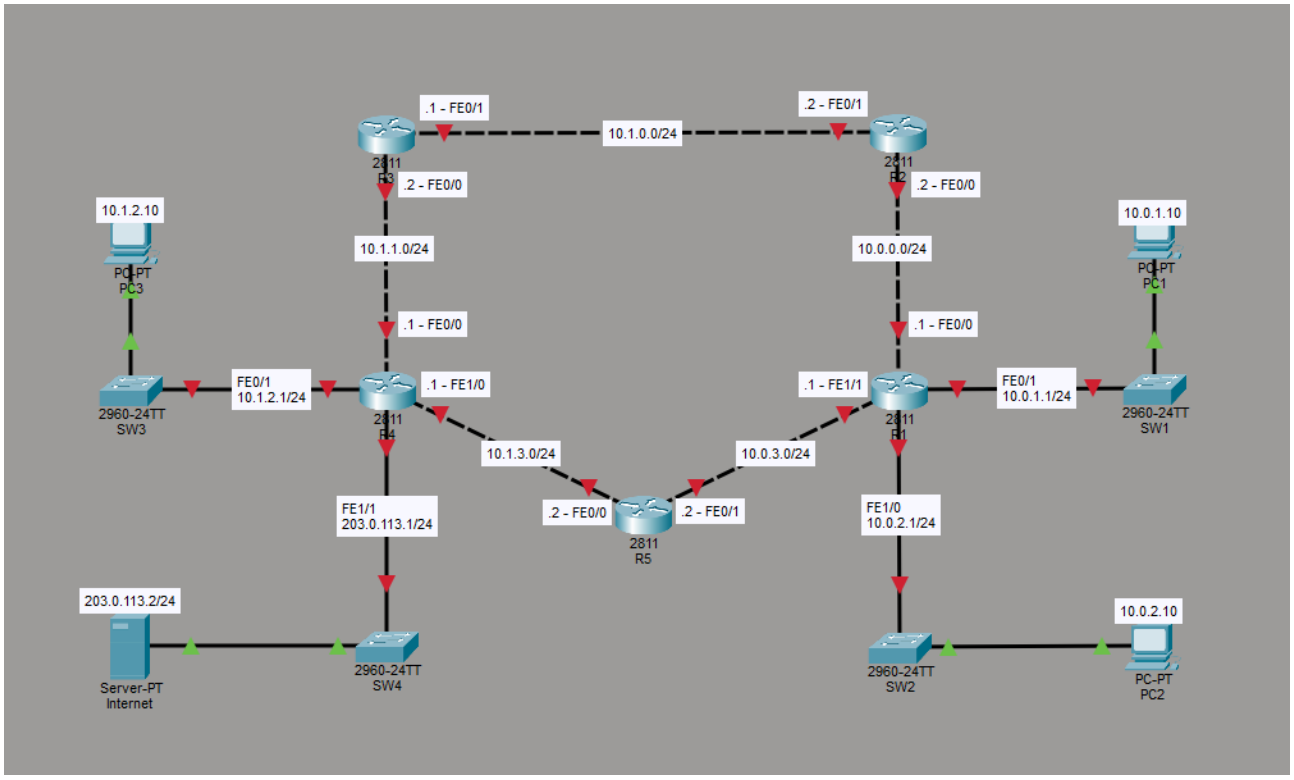
NET512 — Weekly Practical

NAME	Christiaan van Wyk	WEEK	Week 7
STUDENT NO.	20261193	DATE	25 September 2026

Topology

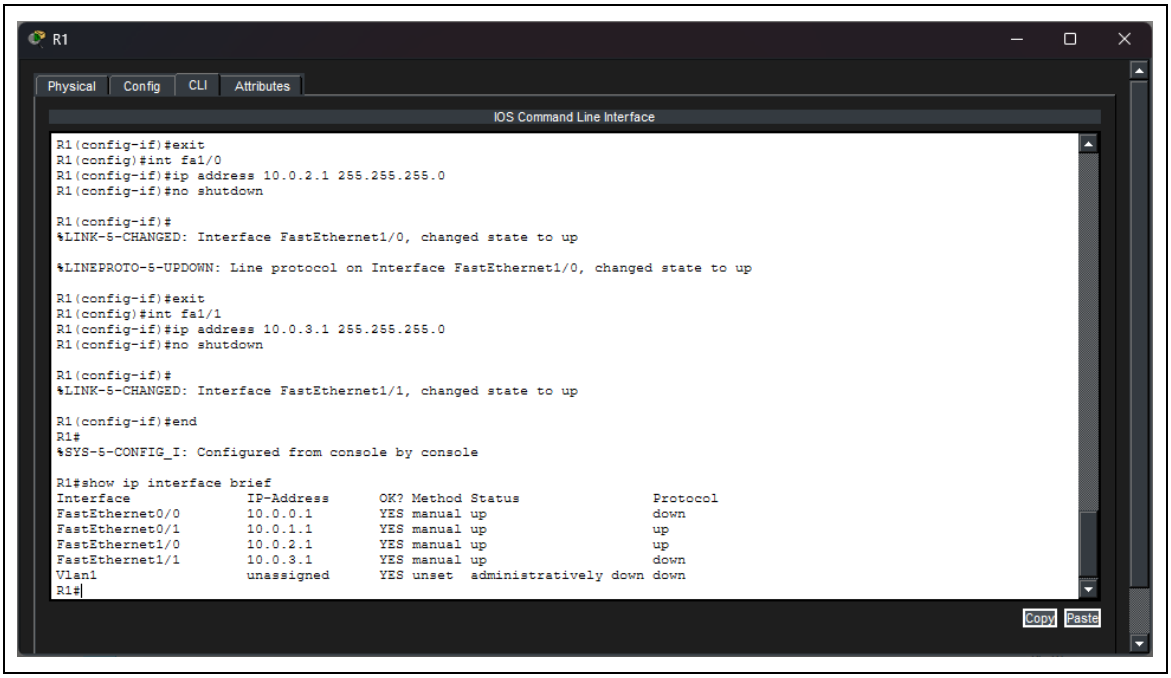


Task 52 — Hostnames

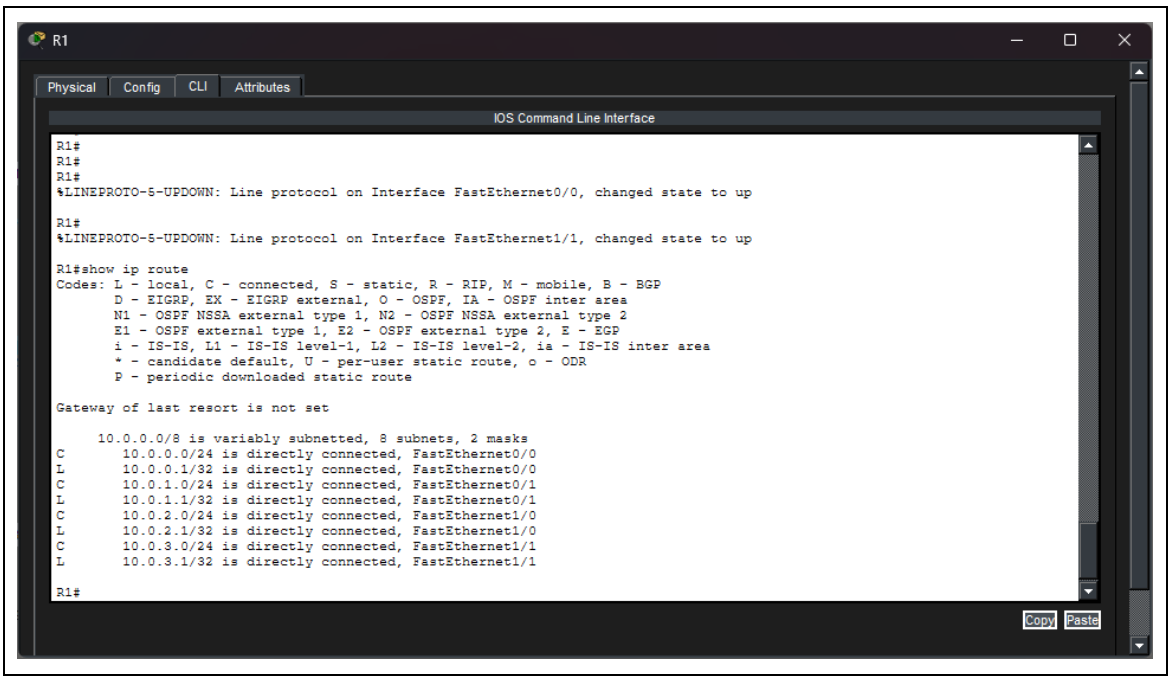



R4
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#conf term
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R4
R4(config)#
Copy
Paste
R5
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#conf term
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R5
R5(config)#
Copy
Paste

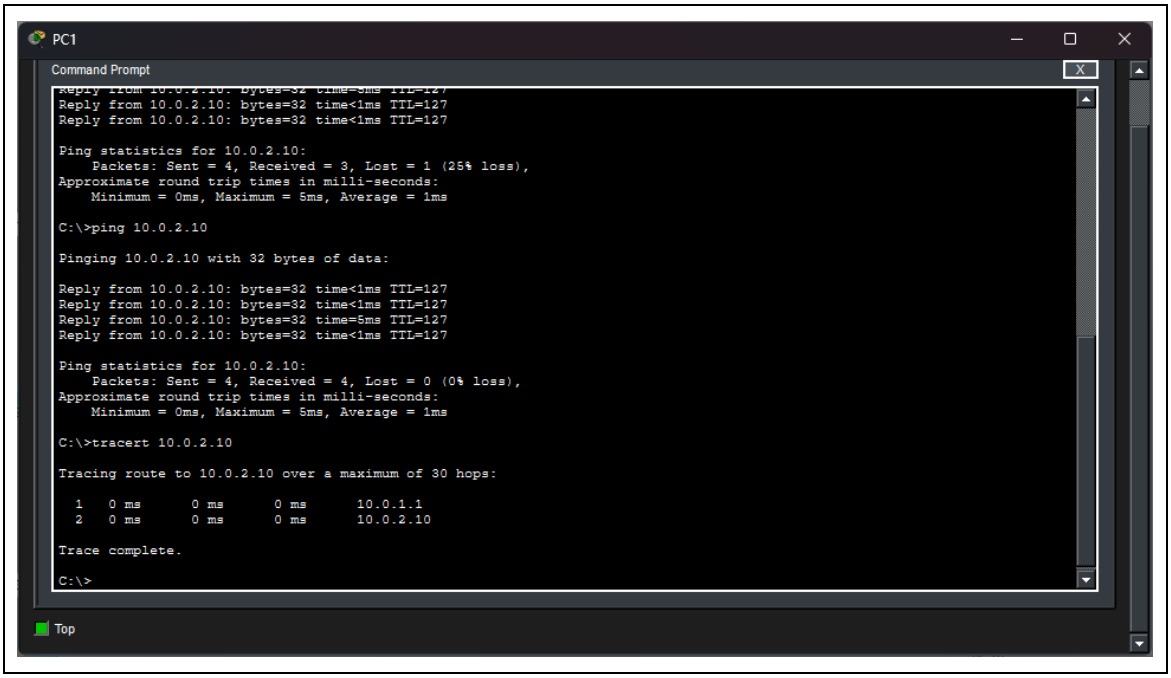
Task 53 — R1 IP Addresses



Task 54 – Verify Connections and Routes



Task 55 — PC Ping and Trace



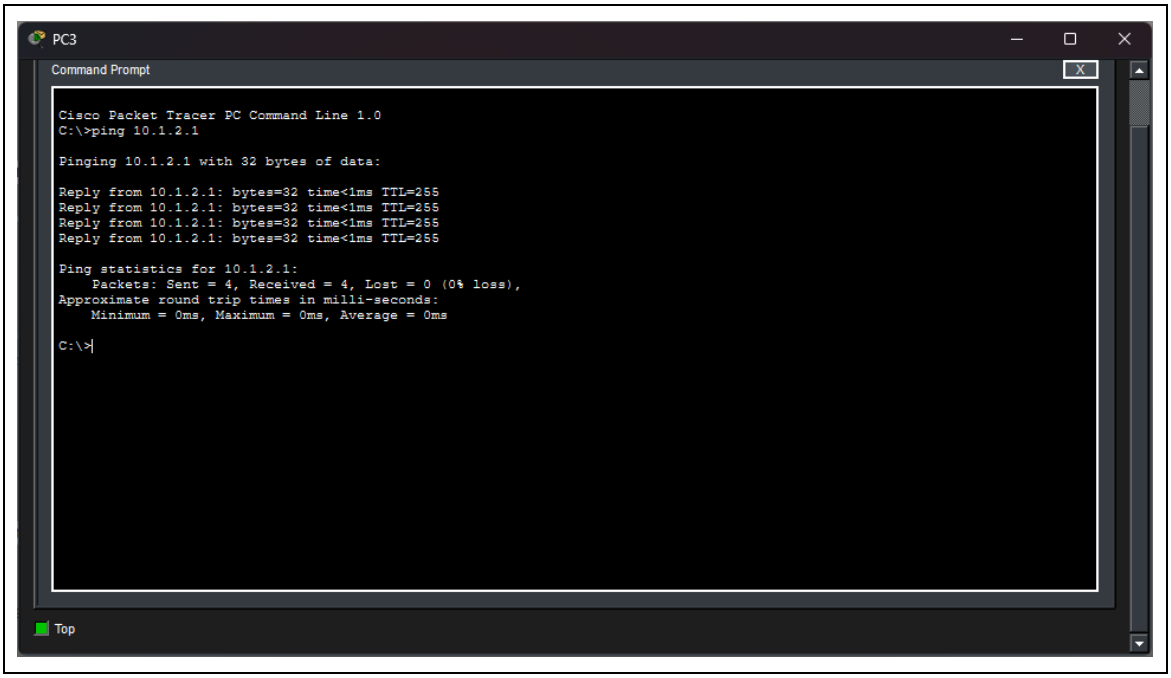
Task 56 — PC1 -> PC3



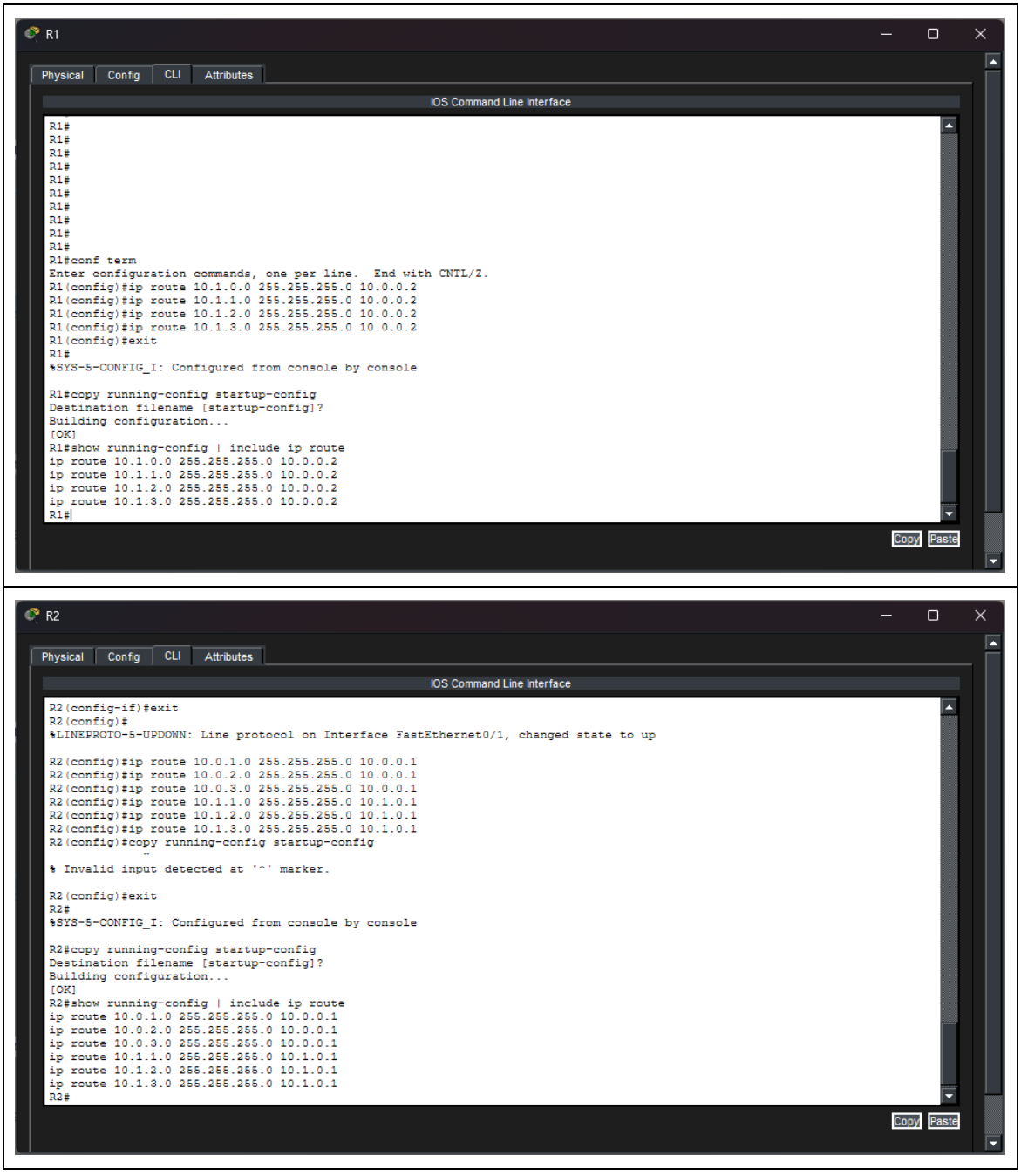
Task 57 — R2, R3, R4 IP Addresses



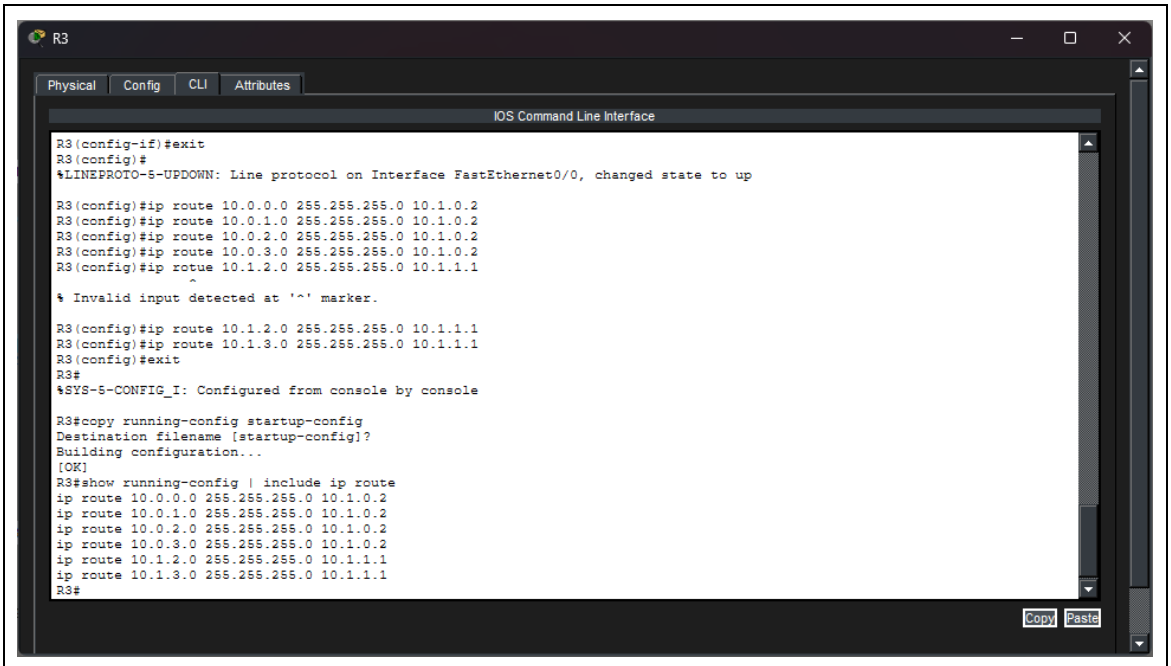
Task 58 — PC3 Default Gateway Ping

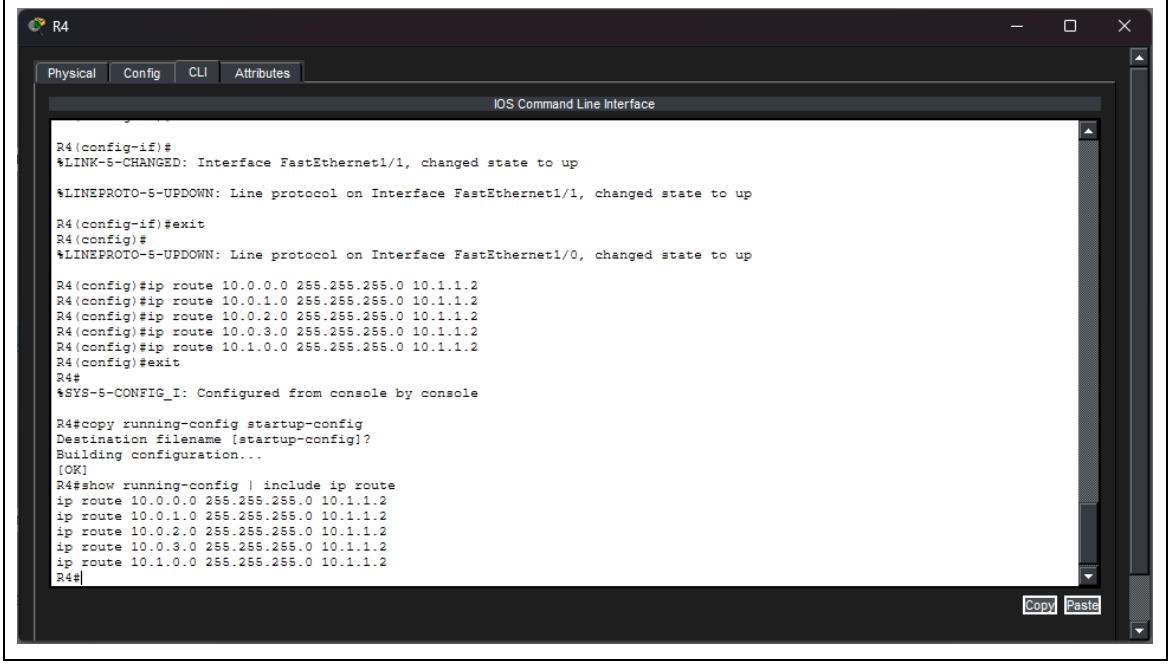


Task 59 — R1-R4 Static Routes

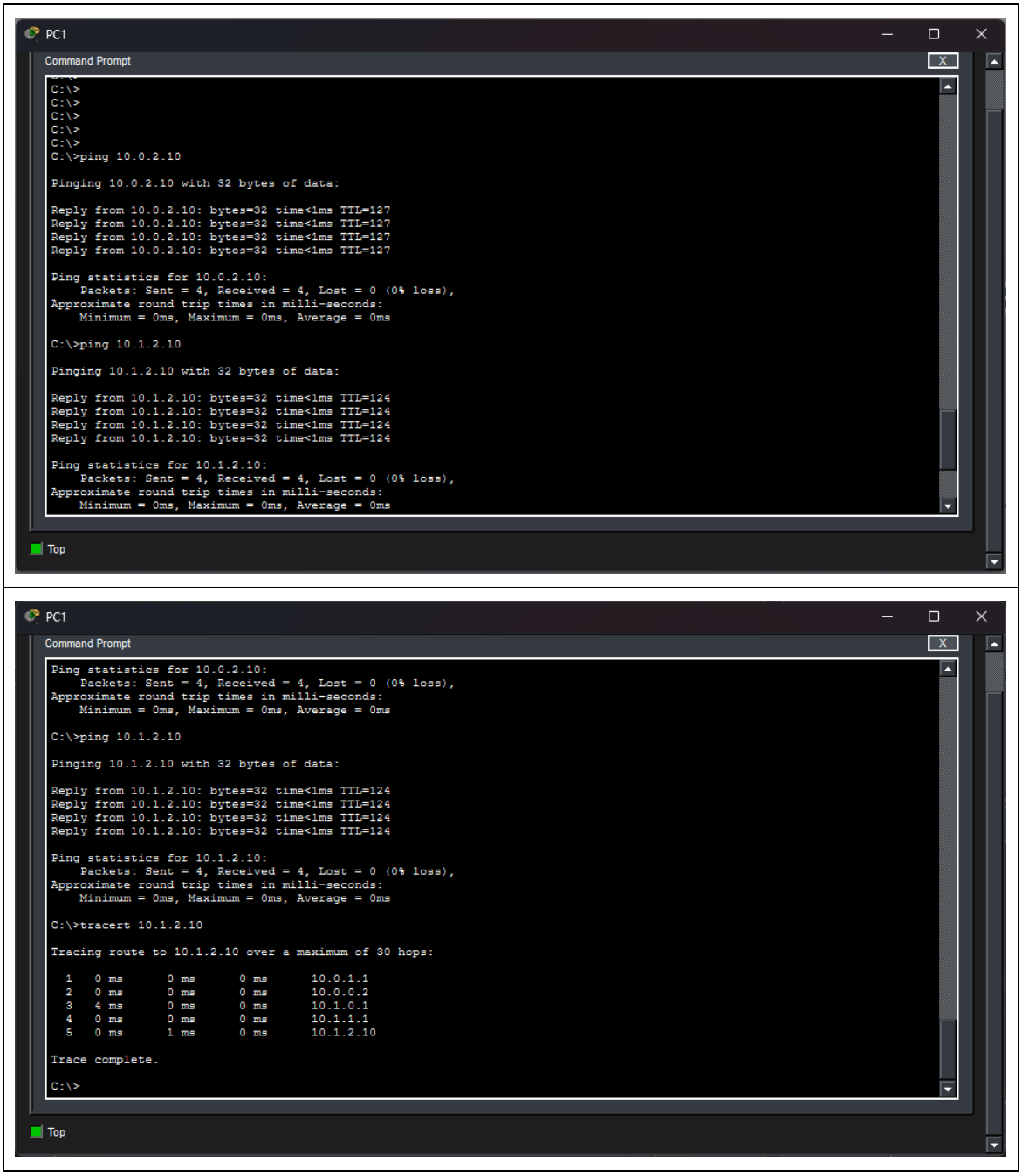


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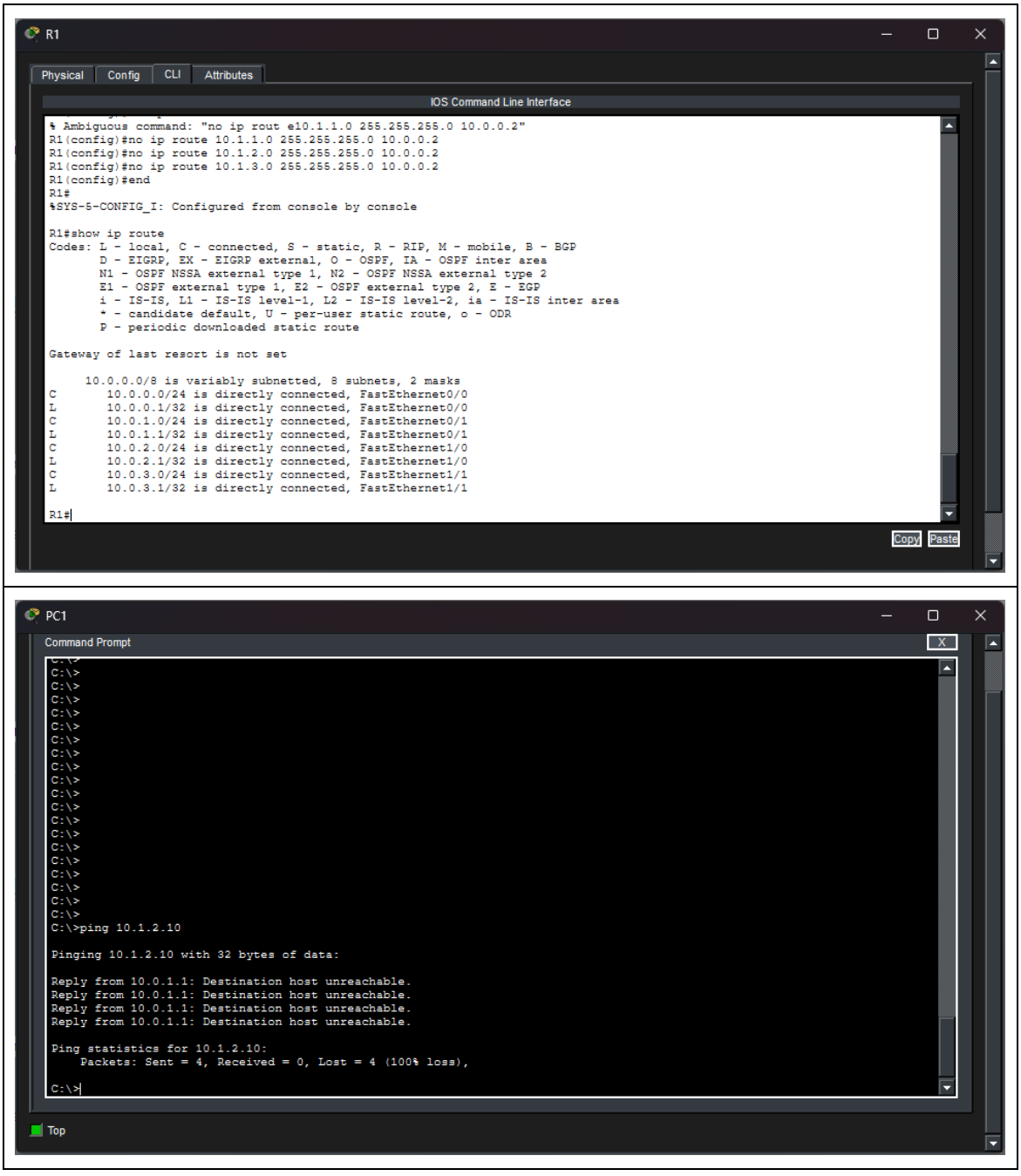

R3
Physical
Config
CLI
Attributes
IOS Command Line Interface
R3(config-if)#exit
R3(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R3(config)#ip route 10.0.0.0 255.255.255.0 10.1.0.2
R3(config)#ip route 10.0.1.0 255.255.255.0 10.1.0.2
R3(config)#ip route 10.0.2.0 255.255.255.0 10.1.0.2
R3(config)#ip route 10.0.3.0 255.255.255.0 10.1.0.2
R3(config)#ip rotue 10.1.2.0 255.255.255.0 10.1.1.1
^
% Invalid input detected at '^' marker.
R3(config)#ip route 10.1.2.0 255.255.255.0 10.1.1.1
R3(config)#ip route 10.1.3.0 255.255.255.0 10.1.1.1
R3(config)#exit
R3#
%SYS-5-CONFIG_I: Configured from console by console
R3#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R3#show running-config | include ip route
ip route 10.0.0.0 255.255.255.0 10.1.0.2
ip route 10.0.1.0 255.255.255.0 10.1.0.2
ip route 10.0.2.0 255.255.255.0 10.1.0.2
ip route 10.0.3.0 255.255.255.0 10.1.0.2
ip route 10.1.2.0 255.255.255.0 10.1.1.1
ip route 10.1.3.0 255.255.255.0 10.1.1.1
R3#
Copy
Paste


R4
Physical
Config
CLI
Attributes
IOS Command Line Interface
R4(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/1, changed state to up
R4(config-if)#exit
R4(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
R4(config)#ip route 10.0.0.0 255.255.255.0 10.1.1.2
R4(config)#ip route 10.0.1.0 255.255.255.0 10.1.1.2
R4(config)#ip route 10.0.2.0 255.255.255.0 10.1.1.2
R4(config)#ip route 10.0.3.0 255.255.255.0 10.1.1.2
R4(config)#ip route 10.1.0.0 255.255.255.0 10.1.1.2
R4(config)#exit
R4#
%SYS-5-CONFIG_I: Configured from console by console
R4#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R4#show running-config | include ip route
ip route 10.0.0.0 255.255.255.0 10.1.1.2
ip route 10.0.1.0 255.255.255.0 10.1.1.2
ip route 10.0.2.0 255.255.255.0 10.1.1.2
ip route 10.0.3.0 255.255.255.0 10.1.1.2
ip route 10.1.0.0 255.255.255.0 10.1.1.2
R4#
Copy
Paste

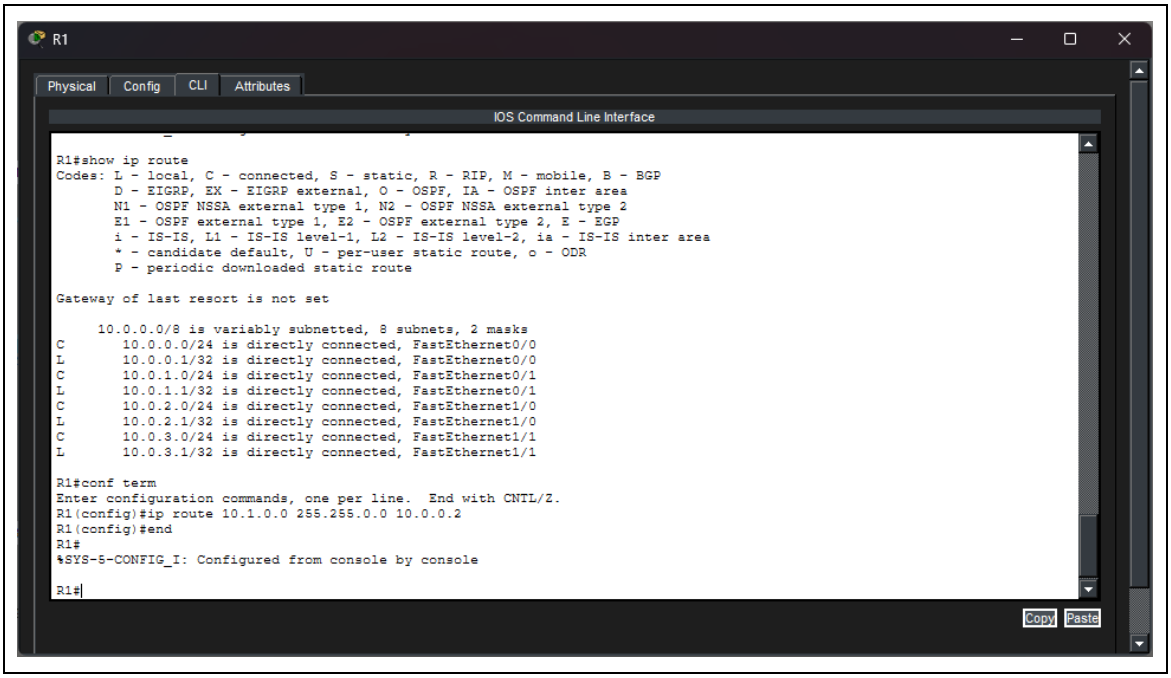
Task 60 — Connectivity & Path Verification



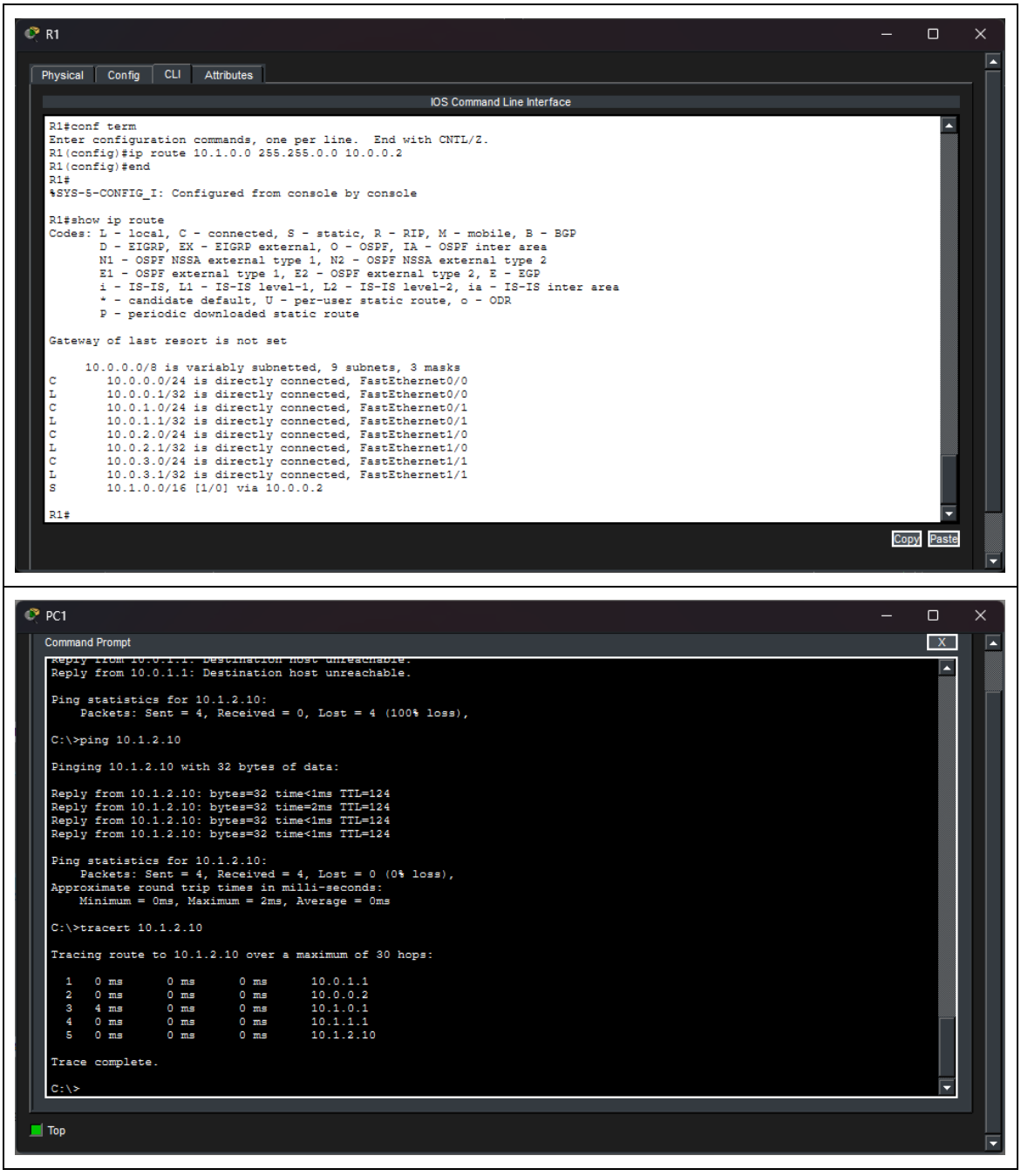
Task 61 — Break R1 Routes



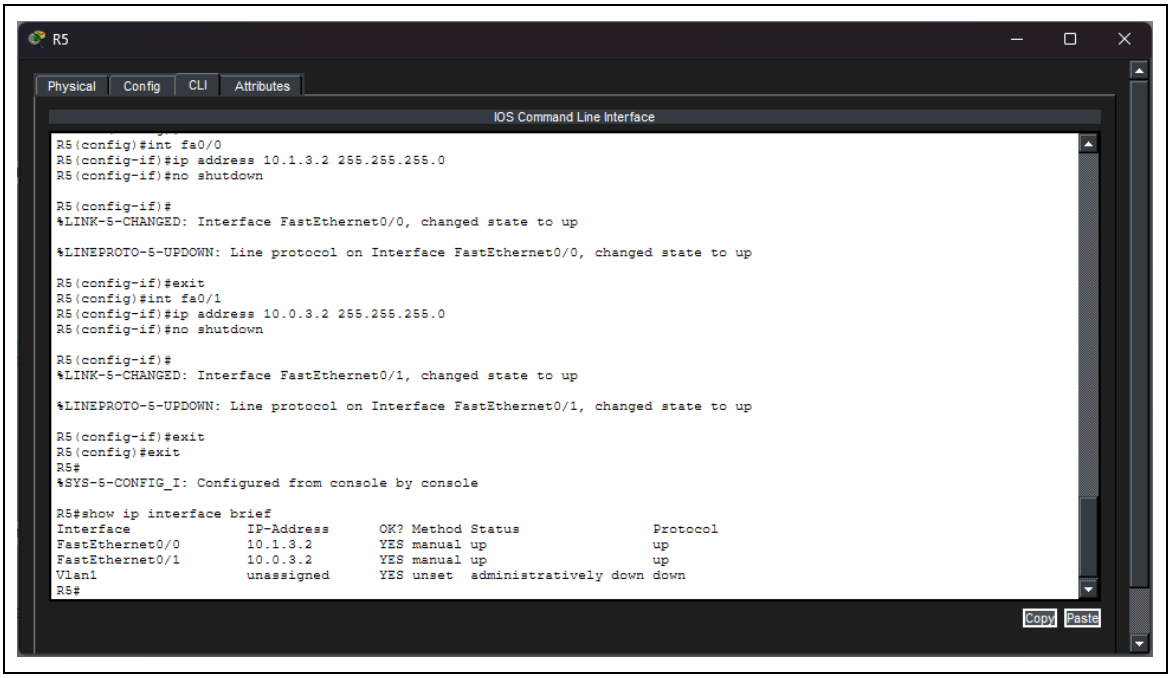
Task 62 — Restore with Single Summary Route



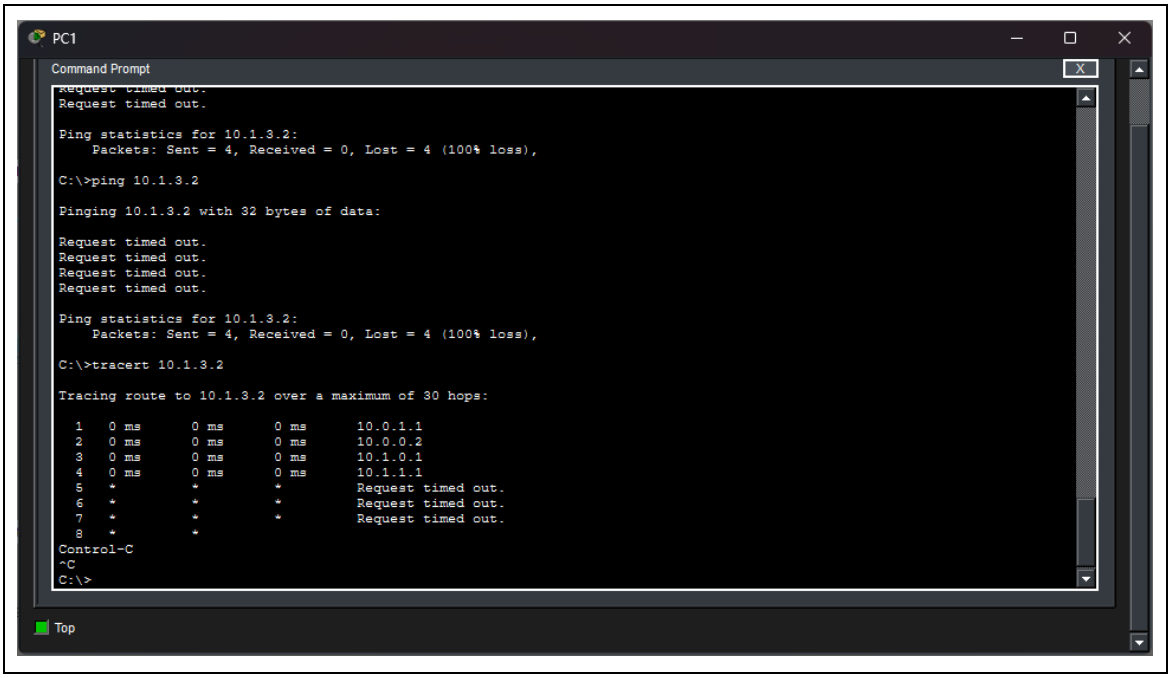
Task 63 — Verify Summary Route



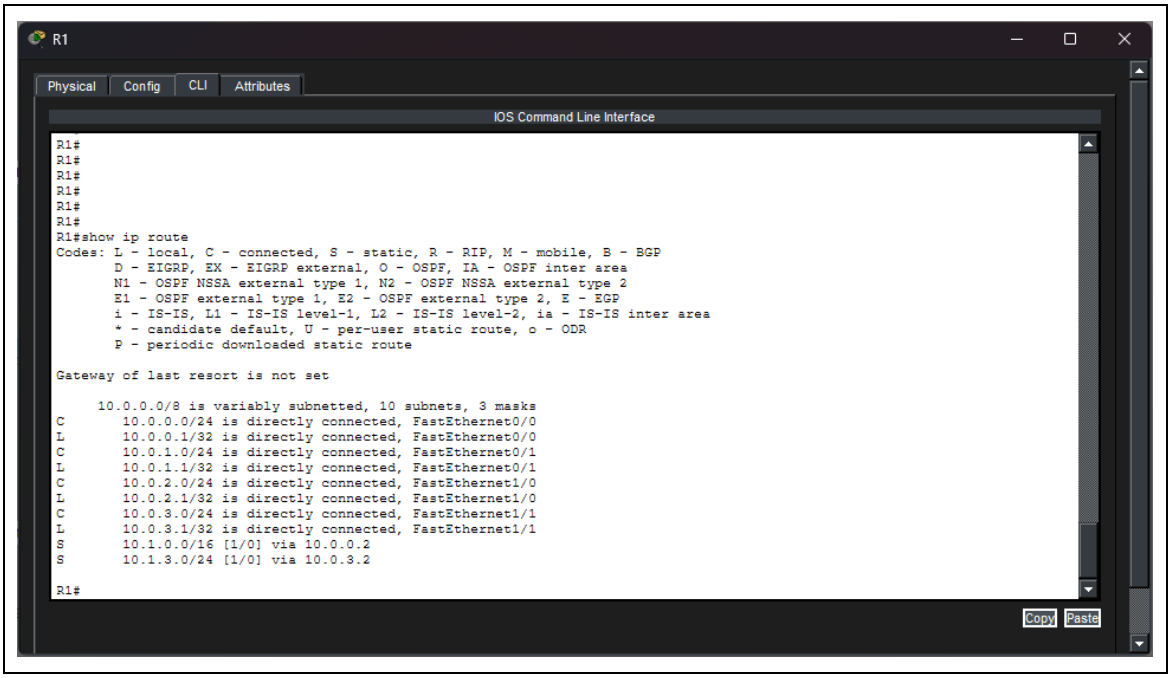
Task 64 — R5 Addresses



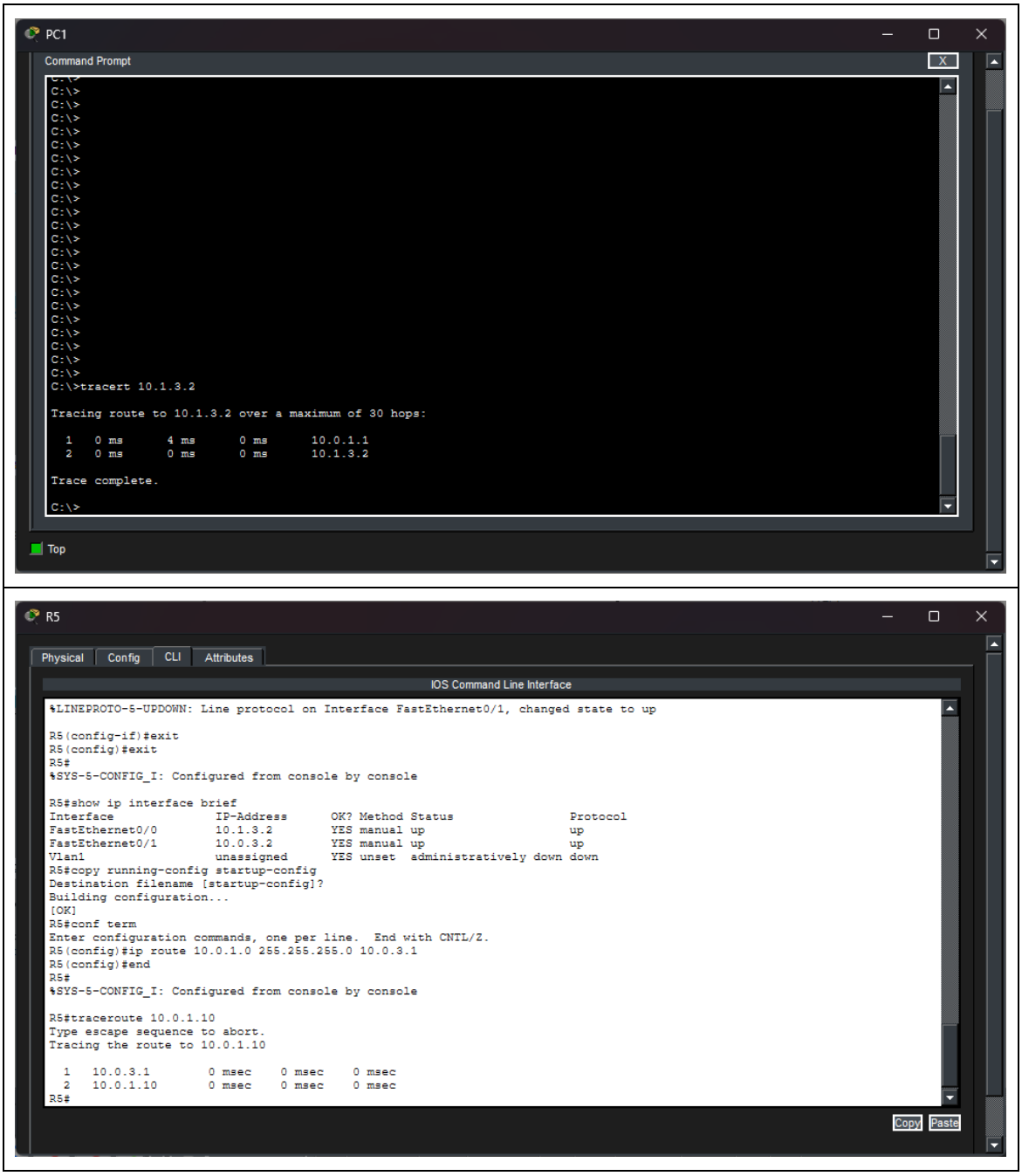
Task 65 — PC1 Reaching R5



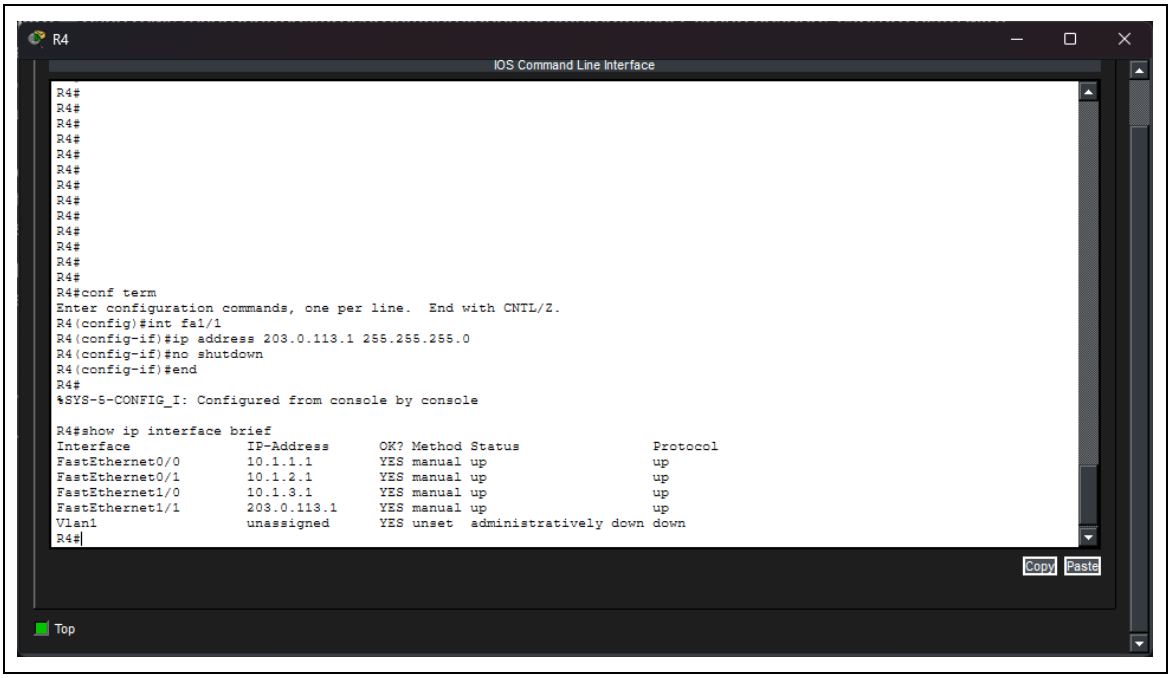
Task 66 — Shortest Path



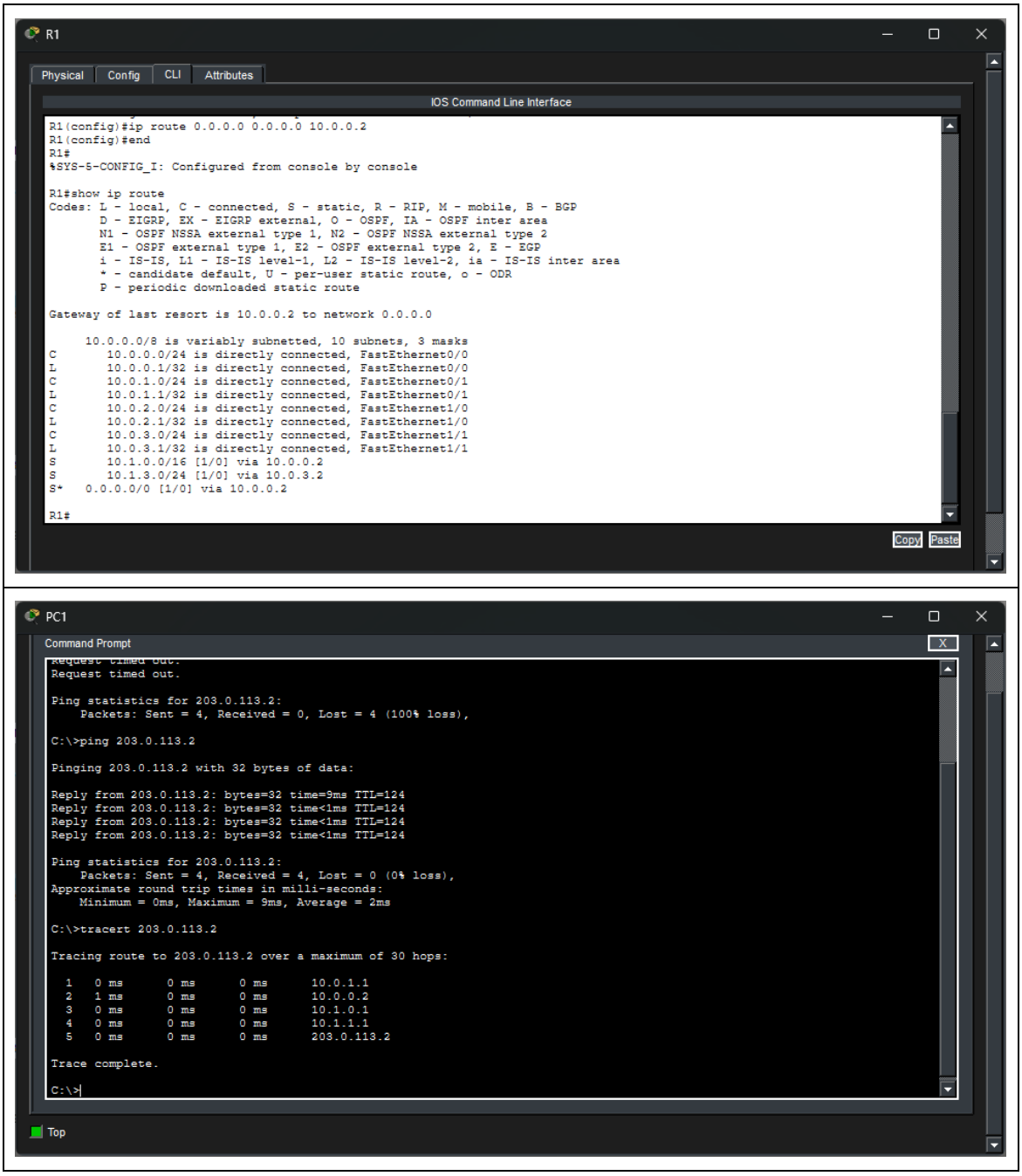
Task 67 — Verify forward/return paths



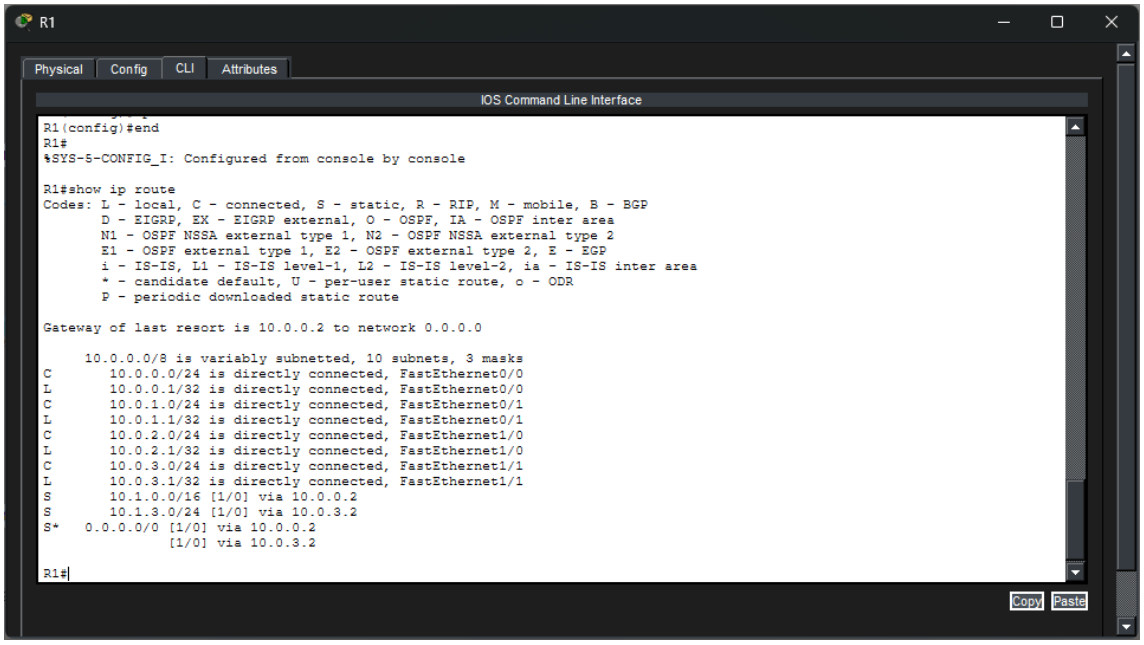
Task 68 — R4 Internet Interface

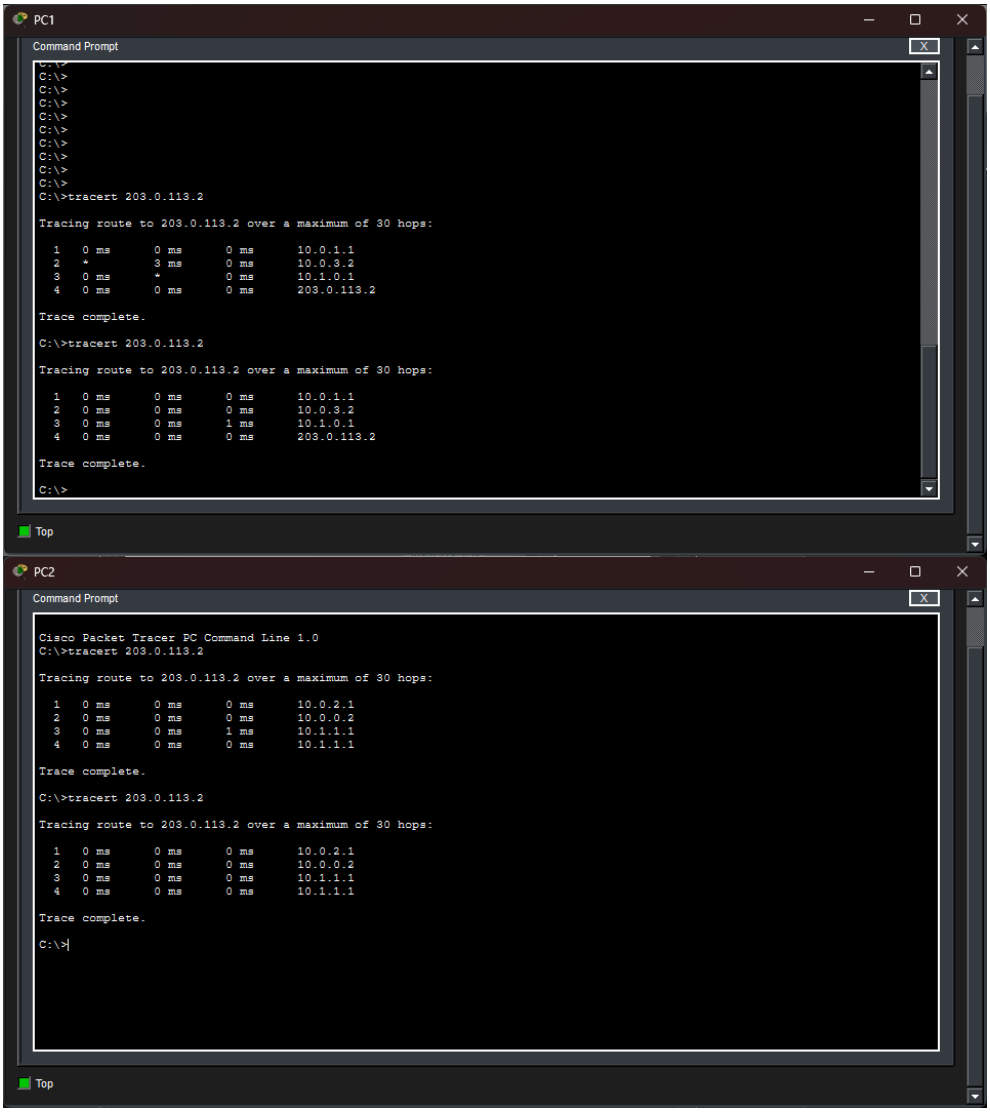


Task 69 — Internet default route



Task 70 — Load Balancing





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20261193

Reflection Questions

Task 71 – Direct Routes

Yes, routes from R1 to every directed network is visible. Using the show route command (Task 70) it shows you all connected Routes (Category C), there are 4 of them:

- 10.0.0.0/24
- 10.0.1.0/24
- 10.0.2.0/24
- 10.0.3.0/24

In addition to the L routes for its own interface addresses on those networks. The routes are visible because they are automatically installed when an interface has obtained an ip address and is up/up.

Task 72 – Reachability without static Routes

PC1 loses reachability due neither pc making the forwarding decision. That decision is done by the routers and in this case the traffic ends at the first one. PC1 has a default gateway of 10.0.1.1 (R1) which is where it sends any non-local traffic. When R1 does a lookup for PC3 it has no static route and thus no matching entry, there is no valid next hop and thus the packet is dropped.

Task 73 – Summary Route

The summary route is able to replace the other static routes because the leading bits of all the other networks are shared. They all start with the same block of bits, so the shorter prefix only covering the leading bits is able to cover all networks at once. The first 2 octets of each network starts with 10.1.x.x and by setting the mask to 255.255.0.0 we cover all the common bits between the networks.